RK520-01 Soil Moisture, Temperature Sensor

QC PASS 01

RK520-01 Soil Moisture, Temperature Sensor is integrated the moisture,temperature measurement.The stainless steel probe is inserted into soil surface or soil profile to test quickly. The product with temperature compensation to ensure the accuracy of measurement.The probe can be permanently embedded underground and be connected to a data logger for unlimited testing.

FEATURES

- High accuracy
- Connect directly to data collector or instrument
- Fully sealed structure
- Placed in any depth underground(5000mm,max.)
- Low power design
- Easy Installation

SPECIFICATIONS

Item	Technical Specification	
	Moisture	Temperature
Range	0-100% （m³/m³）	-30℃-+70℃
Accuracy	±3%(0-50%)	±0.5℃
Output Signal	RS485,0-2V	
Response Time	<1s	
Supply	Mark on the label	
Effective measurement	With the center of the probe diameter is 70mm, high 70mm cylinder	
Housing	ABS	
Dimensions	45*15*145mm(probe:3* Ø3*70mm)	
Operating Temperature	-40℃-+80℃	
Ingress Protection	IP68	
Storage	10-60℃@20%-90%RH	
Probe material	316L stainless steel	

MOUNTING

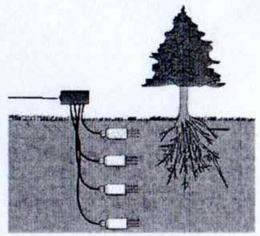
1.Testing medium should be with uniform intensity.

2. If surface soil water content measurement, the sensor should be insert into soil vertically. Do not shake the sensor when inserted, otherwise the probe will be blended;

3. If multi-layer soil water content measurement, the sensor should be buried in the soil and parallel to the ground. Make sure the probe not be blended;

4. When removing sensors from the soil, please hold the sensor housing shell and do not forcibly pull on the cable. Soil on probes should be brushed tightly.

5. Please keep the sensor in dry & clean conditions.



ELECTRICAL CONNECTIONS

Connector（cable）	RS485	Voltage
Red	V+	V+
Black	V-	V-
Yellow	RS485A	
Green	RS485B	Humi.
Brown		Temp

Note: This product has been tested and complies with European CE requirements for EMC directive.

WARRANTY

This product is warranted to be free of defects in materials and construction for a period of 12 months from date of lead time.

Liability is limited to repair or replacement of defective item.

Communication Protocol（MODBUS）

Transmission mode: MODBUS-RTU,**Baud rate:** 9600bps,**Data bits:**8,**Stop bit:**1,**Check bit:**no

Slave address:the factory default is 01H (set according to the need,00H to FFH)

- **The 03H Function Code Example: Read The Moisture, Temperature Value**

Host Scan Order(slave address:0x01)

01 03 00 00 00 02 C40B

Slave Response

01 03 04 0A DC 27 10 222D

Temperature: (0ADC)H=(2780)D,2780/100=27.8℃

Moisture:(2710)H=(10000)D,10000/100=100%

- **The 06H Function Code Example: Modify the slave address(After the restart sensor changes to take effect)**

Host Scan Order (Changed to 02H):

01 06 02 00 00 02 09B3

Slave Response

01 06 02 00 00 02 09B3

If the data≧0x8000, for example:0xFF05,according to the following method to calculate:

0xFF05-0xFFFF-0x01=(65285)D-(65535)D-(1)D=(-251)D,-251/10=-25.1(℃)

If you forget the original address, you should use the broadcast address(FEH) (ensure that no other devices on the bus at this time).

Note:

1. All underlined is fixed bit;

2. The last two bytes is CRC check command.

WARRANTY

This product is warranted to be free of defects in materials and construction for a period of 12 months from date of lead time.Liability is limited to repair or replacement of defective item.